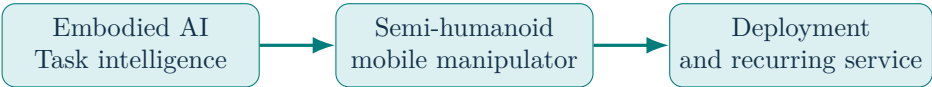


MARKET INTELLIGENCE REPORT

Semi-Humanoid Robotics

Unified Market Analysis: Hardware, High-Level Software and Real-Time Control



Purpose	Investment, product and go-to-market decision support
Scope	Wheeled semi-humanoids / autonomous mobile manipulators; software from cloud to firmware
Prepared	August 2026
Evidence base	Supplied market-analysis dossiers and cited primary/vendor/market sources

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1 Executive view

Investment thesis. Semi-humanoid robotics is best understood as the commercially pragmatic intersection of an autonomous mobile robot (AMR), a collaborative manipulator, and an embodied-AI data platform. A wheeled base avoids the energy, fall-risk and whole-body-control burden of bipedal locomotion while bimanual upper-body kinematics preserve human-environment reach and interaction. The category is therefore positioned to automate material movement plus the manipulation steps that historically required people.

Market signal. Published estimates are not directly comparable because they draw different boundaries around AMRs, mobile manipulators, service robots, systems integration and software. The narrowest supplied estimate, autonomous mobile manipulator robots (AMMRs), grows from \$385.9m in 2023 to \$1.699bn in 2030 (23.9% CAGR). A focused mobile-manipulator estimate reaches \$3.95bn in 2032 from \$1.52bn in 2025 (14.7% CAGR). These figures support a strong-growth conclusion, but *must be used as scenario anchors rather than summed into one market size.*[2, 1]

Strategic implication. The defensible product wedge is not “general humanoid.” It is a repeatable, bounded workflow with measurable throughput and low commissioning friction: intralogistics line-feeding, laboratory material handling, or light industrial kitting. Differentiation should compound across three layers: safe, serviceable hardware; reliable autonomy; and a data/operations layer that improves deployed fleets.



2 Category definition and strategic boundaries

This report uses **semi-humanoid** to mean a wheeled mobile manipulator with an anthropomorphic upper body—typically one or two articulated arms, a perception head and compliant end effectors. It includes AMMRs and research/service mobile manipulators where the system can navigate and manipulate. It excludes stationary cobots, pure AMRs and full bipedal humanoids unless used as a reference class.

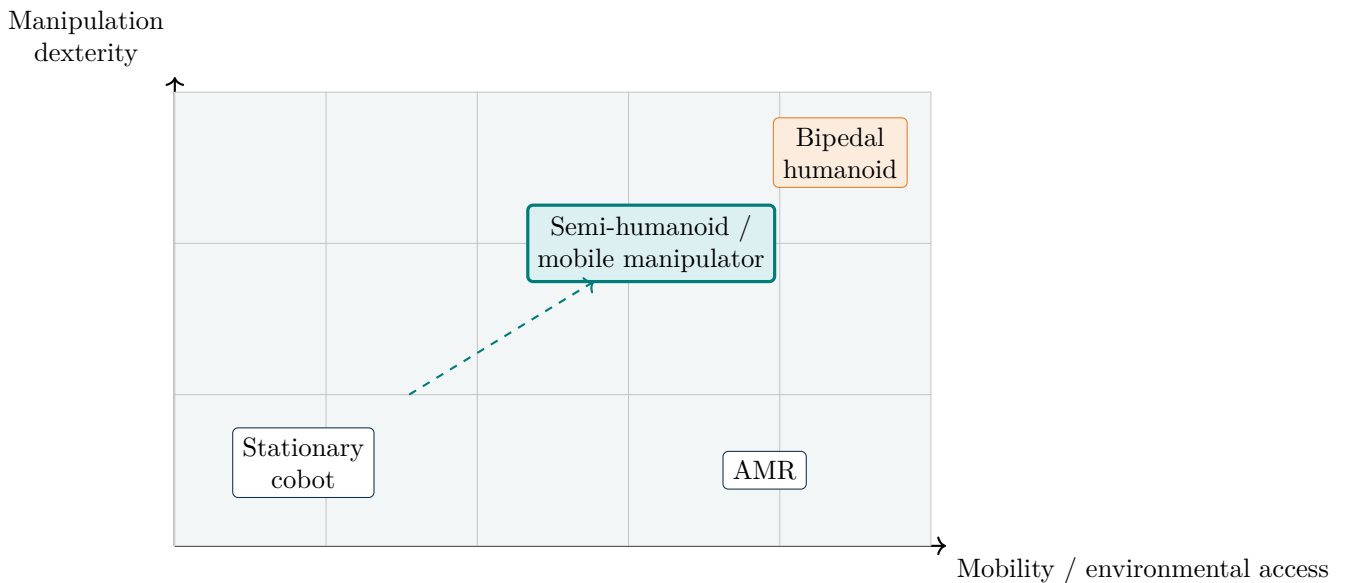


Figure 1: Category positioning: semi-humanoids target the high-value overlap of mobility and dexterous manipulation without requiring bipedal locomotion.

2.1 Why the wheeled form factor matters

For a static support polygon $\mathcal{P}$, a basic feasibility condition is that the ground projection of the system centre of mass remains inside the polygon:

$$\Pi_{xy}(\mathbf{r}_{\text{CoM}}) \in \mathcal{P}, \quad \mathbf{r}_{\text{CoM}}(\mathbf{q}) = \frac{m_b \mathbf{r}_b + m_t \mathbf{r}_t + \sum_{i=1}^n m_i \mathbf{r}_i(\mathbf{q})}{m_b + m_t + \sum_{i=1}^n m_i}. \quad (1)$$

For a wheeled platform, this constraint can be managed through footprint, mass distribution, arm acceleration limits and payload envelopes. A biped must additionally sustain dynamically stable gait and fall recovery. This does not make wheels universally better—stairs remain a clear limitation—but it improves readiness for flat-floor industrial and institutional environments.

Table 1: Form-factor trade-off relevant to near-term deployments.

Dimension	Wheeled semi-humanoid	Bipedal humanoid
Continuous runtime	Typically up to an 8-hour shift-class target in benchmark platforms	Often constrained by dynamic locomotion energy demand
Control burden	Navigation plus manipulation; stable base simplifies planning	Whole-body gait, balance and fall recovery
Facility access	Excellent on flat floors, ramps and standard doorways	Potentially superior on stairs and discontinuous terrain
Commercial fit today	Bounded industrial, logistics, lab and service workflows	Longer-horizon general access and infrastructure adaptation

Source: Synthesis of supplied analyses; platform examples from [7, 8].

3 Market opportunity: size, growth and adoption

3.1 A range-based market model

Market research providers employ different units of analysis. The useful reading is a nested set: AMMR is the narrow product category; mobile manipulation is a broader product category; mobile robots includes adjacent vehicle, software and service revenue. Equation 2 provides the consistency check for every published forecast:

$$\text{CAGR} = \left(\frac{V_T}{V_0} \right)^{\frac{1}{T}} - 1. \quad (2)$$

Table 2: Published market anchors—comparable only within their stated scope.

Scope	Base	Year	Forecast	Year	CAGR / interpretation
AMMR (narrow)	\$0.386bn	2023	\$1.699bn	2030	23.9%; integrated mobile manipulators
Mobile manipulator	\$1.52bn	2025	\$3.95bn	2032	14.7%; focused category estimate
Mobile manipulator (broad)	\$0.89bn	2025	\$3.74bn	2034	17.3%; broader report definition
Extended mobile manipulator	\$4.70bn	2025	\$14.10bn	2033	13.0%; wider systems boundary
AMR baseline	\$3.10bn	2025	\$17.00bn	2035	19.5%; adjacent mobility opportunity
Mobile robots total	\$20.79bn	2025	\$63.28bn	2035	11.83%; includes broad systems/services

Source: Supplied market-analysis dossiers, citing [2, 1, 3, 4, 5, 6]. Currency values are nominal published estimates.

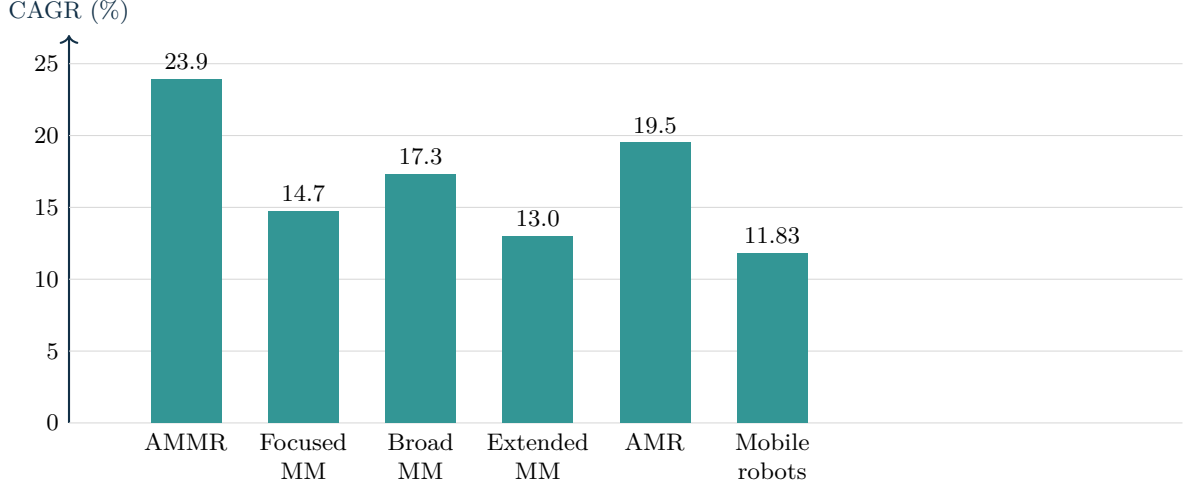
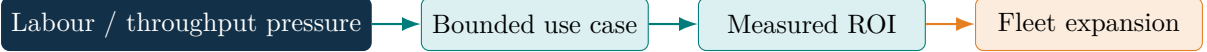


Figure 2: Growth rates reported across non-identical scopes. The category is attractive; a single “market size” is not justified by these sources.

3.2 Demand drivers and buying centres

The primary demand drivers are labour scarcity, throughput pressure in e-commerce and manufacturing, safety/ergonomics, and the need to automate variable tasks without fixed conveyor infrastructure. The first buyers are operations leaders with a quantifiable bottleneck, supported by IT/OT, safety and finance stakeholders. North America is a leading revenue region in the supplied sources, while Asia-Pacific is repeatedly identified as the fastest-growing adoption geography; both signals should be validated per target vertical and country before commercial commitment.[3, 2]



4 Competitive landscape and product economics

The market spans three archetypes: low-cost telescoping single-arm systems, bimanual semi-humanoids and heavy industrial mobile manipulators. The strategic gap is a system that makes bimanual or compliant manipulation deployable without importing the cost and commissioning profile of bespoke industrial integration.

4.1 TCO and procurement logic

The relevant buyer calculation is installed economics, not robot price. A transparent model separates acquisition, integration, operations and residual value:

$$TCO_N = C_{\text{robot}} + C_{\text{integration}} + \sum_{t=1}^N \frac{C_{\text{service},t} + C_{\text{energy},t} + C_{\text{ops},t}}{(1+r)^t} - \frac{V_{\text{residual}}}{(1+r)^N}. \quad (3)$$

The investment case becomes positive when discounted labour, throughput, quality and safety benefits exceed TCO_N . In practice, integration time, exception handling and uptime dominate the uncertainty. The product roadmap should therefore make site mapping, WMS/ERP connectors, safety validation and recovery behaviours reusable assets.

Table 3: Selected benchmark platforms (published specifications; configuration-dependent).

Platform	Archetype	Price (USD)	Mass	Arm payload	Commercial reading
Hello Robot Stretch 3/4	Telescoping single arm	19,950–29,950	46kg	2.5kg extended	Low-cost, narrow-space mobile manipulation
Pollen Reachy 2	Bimanual semi-humanoid	70,000–78,000	50kg	~3kg / arm	Data-rich research and HRI platform
PAL TIAGo	Industrial/research mobile manipulator	~56,000–70,000	70kg	3–6kg	Mature ROS ecosystem, configurable
Toyota HSR	Service/research mobile manipulator	Proprietary / lease	37kg	~1.2kg	Assistive-research benchmark
Willow Garage PR2	Legacy bimanual baseline	~400,000	220kg	~2kg / arm	Demonstrates historical cost/weight burden

Source: Supplied analyses, citing [7, 8, 9]. Prices are list/indicative values, not installed-system quotes.

5 The software market: where value and defensibility accrue

Semi-humanoid software is not one market; it is a value chain from embedded control to cloud operations. High-level software converts intent and observations into plans. Low-level software makes physical execution deterministic, safe and observable. The commercial opportunity is recurring when the supplier owns deployment tooling, fleet operations, updates, analytics and workflow integrations—not merely the robot application.

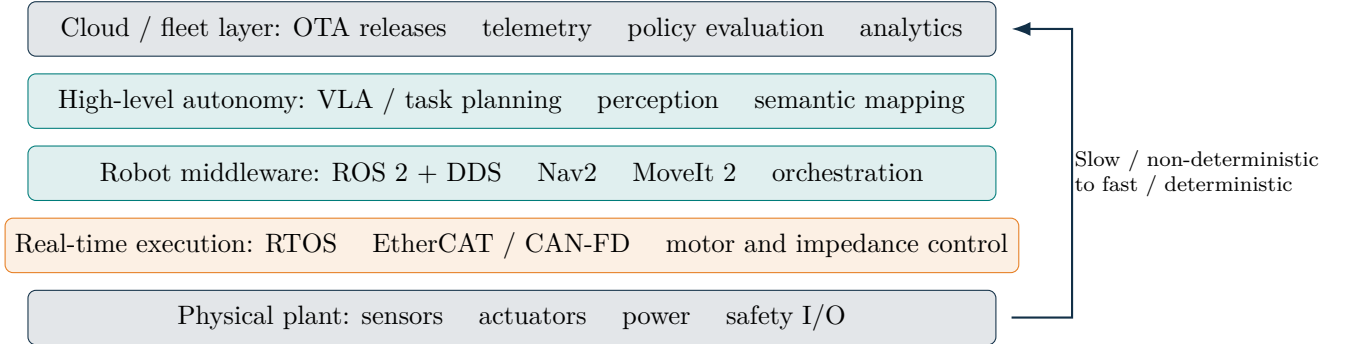


Figure 3: Unified software architecture. Separation of decision-making from real-time safety control is a design requirement, not an implementation preference.

5.1 High-level software

At the high level, ROS 2 and DDS provide modular communications; Nav2 handles navigation; MoveIt 2 produces manipulation plans; perception builds scene understanding; and VLA or task models translate human intent into constrained action sequences. Simulation (Isaac Sim/Isaac Lab, MuJoCo and Gazebo) accelerates validation, synthetic data and policy learning.[10, 11, 12, 13]

The attractive software economics lie in workflow packs and operations: site configuration, reusable behaviours, remote diagnostics, secure OTA deployment, teleoperation assistance and fleet analytics.

Generic foundation models may lower the cost of cognition, but production-grade reliability, data governance and site integration remain local moats.

5.2 Low-level software and safety

Low-level control belongs on deterministic compute, independent of an AI planner. A typical cascade is:

$$\tau_{\text{cmd}} = K_p(q_d - q) + K_d(\dot{q}_d - \dot{q}) + \tau_{\text{ff}}, \quad (4)$$

$$\mathbf{F}_{\text{cmd}} = \mathbf{K}(\mathbf{x}_d - \mathbf{x}) + \mathbf{D}(\dot{\mathbf{x}}_d - \dot{\mathbf{x}}), \quad (5)$$

where q is joint state and $\mathbf{x}$ is task-space pose. The firmware enforces torque, velocity, temperature and communication-loss limits before commands reach the actuator. EtherCAT is attractive where high-rate synchronisation is needed; CAN-FD provides a robust, lower-complexity alternative for distributed subsystems.[8]

Table 4: High-level versus low-level software: product and engineering roles.

Aspect	High-level autonomy	Low-level execution
Primary goal	Choose and plan useful work	Execute safely and predictably
Typical tools	ROS 2, DDS, Nav2, MoveIt 2, VLA/VLM, simulation	RTOS, C/C++/Rust, fieldbus, motor drivers, safety watchdogs
Time scale	Tens of milliseconds to seconds	Sub-millisecond to millisecond-class loops
Failure posture	Degrade, request help, re-plan	Fail safe, limit energy, maintain deterministic state
Commercial lever	Workflow packs, data, fleet operations	Reliability, certification evidence, hardware differentiation

6 Data flywheel and sim-to-real

Teleoperation provides an early route to revenue and a disciplined route to autonomy. Demonstrations create labelled observation-action data; simulation expands rare conditions; evaluation gates protect deployment; fleet telemetry identifies exceptions. Domain randomisation varies lighting, textures, friction, mass and latency so policies avoid overfitting to a simulator.[11, 12]

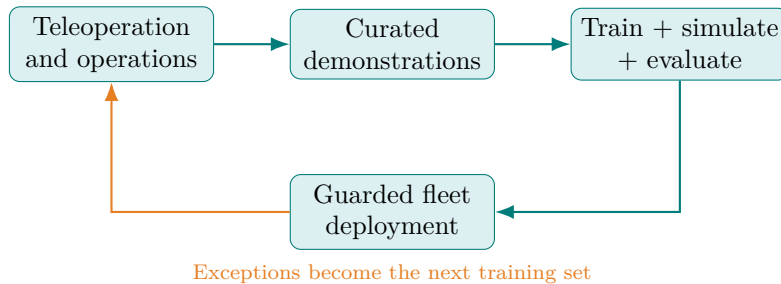


Figure 4: A commercial autonomy flywheel: operational data is valuable only when governed, curated and tied to measurable evaluation gates.

Table 5: Beachhead screen.

Vertical	Initial workflow	Why it fits	Watch-out
Warehouse / intralogistics	Line feeding, tote movement, simple picking	Clear throughput metrics; flat-floor mobility	WMS integration and exception rate
Laboratories / healthcare operations	Specimen, consumable and equipment transport	Repetitive routes; traceability value	Validation, hygiene and liability
Electronics / light manufacturing	Kitting, machine tending, material replenishment	High labour value; controlled environments	Precision and cycle-time requirements
Research / education	Teleoperation and embodied-AI development	Early-adopter appetite; data creation	Smaller budgets; feature-led buying

7 Go-to-market priorities and risks

7.1 Recommended beachheads

7.2 Risk register and mitigation

Risk	Commercial impact	Mitigation
Commissioning is bespoke	Margin erosion; slow sales cycle	Productise connectors, maps, safety templates and acceptance tests
Manipulation exceptions	Low autonomy uptime	Start with constrained objects; use compliant end effectors and remote assist
Safety / cyber exposure	Delayed approvals; reputational risk	Independent safety layer, signed OTA, audit logs and network segmentation
Model unpredictability	Unsafe or unreliable behaviour	Constrain AI outputs to verified skills; simulation and on-robot gates
Hardware service burden	Fleet downtime	Modular actuators, diagnostics and designed-for-service access

8 Recommended product strategy

- 1. Sell a workflow outcome, not a general-purpose robot.** Package one repeatable task with a baseline site survey, integration boundary, safety case and success metric.
- 2. Build the platform around safe modularity.** A wheeled base, compliant upper body, proximal actuation and passive thermal design support serviceability and lower distal inertia. Isolate AI planning from deterministic motor and safety control.
- 3. Make software the scale mechanism.** Invest early in configuration tools, fleet observability, versioned releases, dataset governance and teleoperation. These lower deployment cost and improve margins with fleet size.
- 4. Gate autonomy honestly.** Use a progression from teleoperated assistance to supervised autonomy to measured autonomous execution. Promote only workflows whose recovery rate, safety performance and economics are demonstrated at customer sites.
- 5. Measure the right unit economics.** Track productive hours, interventions per 100 tasks, mean

time to recovery, deployment days, service hours per robot-month and customer payback—alongside hardware gross margin.

9 Conclusion

Semi-humanoid robotics is a credible near-term automation category precisely because it declines to solve every humanoid problem at once. Its opportunity is created by the convergence of mobile base capability, dexterous upper-body manipulation and an increasingly capable software stack. The winning proposition will couple a focused use case with a demonstrably safer real-time control layer and a software operating model that turns every deployment into lower-cost, higher-confidence future deployments.

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